

NADINE EL-MUFTI

Canadian Citizen, residing in 📍 Montreal, QC

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EXECUTIVE SUMMARY

Machine learning researcher and data scientist specializing in healthcare AI, with experience developing end-to-end NLP and speech-processing pipelines for neurological and psychiatric applications. Skilled across the full machine learning lifecycle, including dataset curation, clinical data engineering, statistical and confounding analysis, model development, evaluation, optimization, and interpretability. Active researcher in responsible AI and AI governance, studying how values shape the design and governance of healthcare AI through stakeholder-informed and anticipatory approaches.

WORK EXPERIENCE

The Douglas Research Centre, McGill University

Graduate Research Assistant

Montreal, QC

Jun 2025 – Present

- Driving speech-based relapse prediction research in psychotic disorders in collaboration with Dr. Lena Palaniyappan and Dr. Alban Voppel by maintaining multisite data collection using REDCap workflows and powering the Voici Speech Bank by engineering and analyzing pipelines to clean, align, and structure longitudinal speech data across diverse studies. This work enables multimodal insights into voice, symptoms, and cognition, paving the way for more predictive, compassionate mental health care.

Mila - Quebec AI Institute

Collaborating Researcher / Master's Research

Montreal, QC

Oct 2024 – Present

- Leading research on speech as a biomarker for the detection and longitudinal monitoring of Parkinson's disease, covering the full machine learning lifecycle from dataset curation and clinical data integration to model training, evaluation, and optimization, while investigating confounding factors across in-the-wild and clinical recordings.

Applied Perception Lab, Concordia University

Graduate Research Assistant

Montreal, QC

Dec 2024 – Aug 2026

- Leading research on anticipatory fairness and governance in healthcare AI, deriving a ten-value, SDG-aligned framework from stakeholder deliberation, applying it to evaluate future deployment scenarios with cross-disciplinary raters, and extending it into ethical personas for governance committees.

Concordia University

Teaching Assistant

Montreal, QC

Jan 2025 – April 2025

- Achieving consistent and fair grading for Concordia's Machine Learning course 'Comp 432' by marking labs, delivering prompt feedback, and addressing inquiries from 110 students.

PUBLISHED & ON-GOING RESEARCH

- **El-Mufti, N.**, Mackinley, M., Dzialoszynski, P., Palaniyappan, L., Voppel, A.* Automatic Speech Recognition Performance in Psychiatric Speech: Linguistic, Clinical, and Architectural Factors (**Submitted to NPP—Digital Psychiatry and Neuroscience (DPN), Nature Portfolio**)
- **El-Mufti, N.**, Drouin, S., Jannin, P., Kersten-Oertel, M.* Futures Before Failures: Design Fictions for Anticipatory Fairness in Surgical AI. (**Accepted at the Joint FAIMI-BRIDGE-EPIMI Workshop, MICCAI 2026**)
- **El-Mufti, N.**, Drouin, S., Jannin, P., Kersten-Oertel, M.* VIRTUES: A Design-Fiction Framework for Eliciting the Values of Sustainable, Responsible Surgical Innovation. (**Under review; preprint forthcoming on arXiv**)

- **EI-Mufti, N.**, Kersten-Oertel, M., Ravanelli, M.* Enhancing Speech-Based Parkinson's Detection with Celebs4PD: A Corpus of Natural, In-the-Wild, Longitudinal Celebrity Speech. *(In Preparation)*
- **EI-Mufti, N.**, Kersten-Oertel, M., Glatard, T., Klein, D., Ravanelli, M.* Confounding Factors in Speech Biomarkers of Parkinson's Disease. *(In Preparation)*
- Reyes, J., **EI-Mufti, N.**, Gorman, S., Xie, D., Kersten-Oertel, M.* User-Centered Design for Surgical Innovations: A Ventriculostomy Case Study. EPIMI Workshop, MICCAI 2022, LNCS vol. 13755, 2022. https://doi.org/10.1007/978-3-031-23223-7_5

LEADERSHIP, SERVICE & ENGAGEMENT

[Arabs in Neuroscience](#) (Member, Apr 2026 - Present) · [The Levantines](#) (Member, Feb 2026 - Present) · [Association Étudiante de Mila](#) (Member, Oct 2025 – Present) · [Arbour Foundation Scholars](#) (Member, Sep 2025 – Present) · [Golden Key International Honour Society](#) (Member, Sep 2025 – Present) · [Brainhack MTL](#) | [Brainhack Global](#) (Organizer, Jul 2025 – Present) · [Graduate Students' Association of Concordia GSA](#) (Subcommittee Member, Jan 2025 – Apr 2025) · [School of Graduate Studies, Concordia University](#) (Ambassador, Sep 2024 – May 2026) · [GEMinAI](#) | [Concordia AI2](#) (Mentee, Oct 2024 – May 2025)

EDUCATION

Master of Computer Science Computer Science <i>Concordia University / Mila - Quebec AI Institute</i>	Montreal, Quebec Jan 2025 – Present
Bachelor of Engineering Software Engineering GPA (3.67) With Distinction <i>Concordia University</i>	Montreal, Quebec Jan 2019 – May 2024

HONORS & AWARDS

Power Corporation Of Canada Graduate Fellowship Issued by Concordia University	Jun 2026
Arbour Foundation Master's Scholarship Issued by The Arbour Foundation	Sep 2025
FRQNT B1X Award Competition – 92/100, Ranked 9th (Funding Cutoff: 8) Issued by the Fonds de recherche du Québec	Apr 2025
NSERC CGS-M Award Issued by the Natural Sciences and Engineering Research Council of Canada	Apr 2024
NSERC Undergraduate Student Research Award (USRA) Issued by the Natural Sciences and Engineering Research Council of Canada	May 2022

LICENSES & CERTIFICATIONS

[Transatlantic Summer School](#) (UdeM x LMU, Jun 2026) · [Master AI Tools for Research](#) (WiseUp Communications, Jun 2026) · [Foundations in Responsible AI & AI Ethics](#) (Mila, May 2026) · [Bell x Mila x Buzz HPC x Kids Help Phone Hackathon](#) (Mila, Apr 2026) · [Ethics and Governance of AI for Health](#) (WHO Academy, Jan 2026) · [QcSE Program](#) (V1 Studio, Jun 2025) · [Strategic Public Communications](#) (GradProSkills, Mar 2025) · [TRAIL AI Learning Certificate](#) (Mila, Mar 2025) · [TCPS 2: CORE 2022](#) (Government of Canada, Jan 2025) · [Tech Stewardship Certification](#) (Tech Stewardship Network, Jul 2024)